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## Product Datasheet | **Model 750**

*Mini Pulse & Delay Generator*

# Table of Contents

|                                          |          |
|------------------------------------------|----------|
| <b>1.Overview</b>                        | <b>3</b> |
| Features                                 |          |
| Description                              |          |
| Applications                             |          |
| <b>2.Front &amp; Rear Panel</b>          | <b>4</b> |
| Connectors& Switches                     |          |
| Pulse Shaping Modules                    |          |
| <b>3.Specifications</b>                  | <b>5</b> |
| Options                                  |          |
| Ordering Information                     |          |
| <b>4.Operating Information</b>           | <b>6</b> |
| Block Diagram of the Generator           |          |
| Trigger Controller & Trigger Sources     |          |
| Delay Channel & Channel Output           |          |
| <b>5.Example of Channel Outputs Mode</b> | <b>7</b> |
| Mode Repetitive andSingle                |          |
| Mode Burst and Mux                       |          |
| Pulse Picking                            |          |
| <b>6.Example of Output Pulse</b>         | <b>8</b> |
| <b>7.Control and Software Tools</b>      | <b>9</b> |
| Easy Remote (Web Control Panel)          |          |
| General Remote (Ethernet/USB)            |          |



## Applications

- System Laser Timing Control
- ATE Application
- Laser Pulse Picking
- Precision Pulse Application
- Instrument Triggering
- Components Test

## Features

- 4 independent delay channels (or 8 in option) — 100 ps delay resolution (or 1 ps in option), 50 ps channel-to-channel RMS jitter (4 ps in option)
- Output pulse 1.5 to 5 V into 50  $\Omega$ , 1 ns rise and fall time with independent control of width, polarity, amplitude, burst and MUX mode
- Up to 50 MHz independent trigger rate (repetitive or single) for every channel
- External trigger mode with pre-scaler or internal trigger mode from three synchronous programmable timers or command
- Gate (or second trigger) input
- External clocking up to 240 MHz (user programmable)
- Controlled via USB and Ethernet
- Ultra-compact packaging and low power
- External AC/DC compact power supply

## Description

The Model 750 Mini Pulse & Delay Generator provides 4 (or 8 in option) independent delayed pulses. Delays up to 100 seconds can be programmed with 100 ps resolution and channel-to-channel jitter less than 50 ps RMS. An option enhances delay resolution to 1 ps, with channel-to-channel jitter under 10 ps and narrow-pulse mode down to 3 ns.

SMB outputs deliver 1.5 V to 5 V, 1 ns rise time pulses, into 50  $\Omega$ . Pulse amplitude, polarity, width and burst count are adjustable on each output channel.

The Model 750 offers two inputs or three internal synchronized Timers (adjustable from 0.01 Hz to 50 MHz) or software command for triggering all selected delay channels. Either trigger rate may be set as one-shot or repetitive. Gate Input allows quickly inhibiting all selected channel Outputs: this input function can be selected as a second external Trigger or as an input/output for daisy chaining two Model 750 units to increase channel number.

The generator uses an internal 100 MHz TCXO clock reference, or an external user-programmable (from 10 MHz to 240 MHz) clock (sine or square).

On the front panel, many LED indicators are included to indicate when input or output are running, and a RUN/STOP button enables/disables the output for all channels.

Model 750 parameters can be remote controlled via USB to UART or Ethernet.

A multi-channel “Real Time Chart View” allows the user to visualize every channel output in a precise common time chart on your PC. Zoom and cursor functions allow precise time measurement and control.

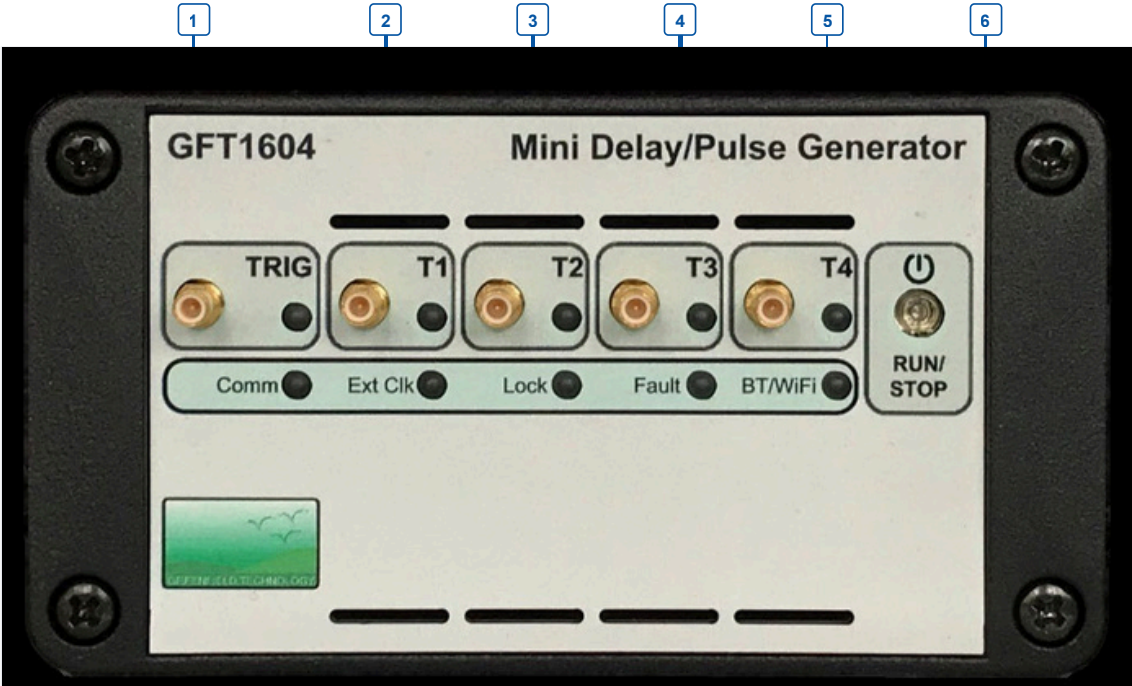
**Application example:** The Model 750 is well suited to synchronize all the devices of a Picosecond Laser System with only one compact unit and one GUI. In this application the “Clock reference input” of the delay generator receives a reference signal (80 MHz for example) from laser oscillator via an O/E (optical to electrical converter).

From the Model 750 Generator each amplifier (Pump-laser, Q-switch, Pockel cell...) or diagnostic instruments (Digitizer, Calorimeter, CCD camera...) can receive repetitive or single pulses (adjusted in rate, delay, amplitude, polarity and width) synchronized on “Clock reference input” with a very low jitter. From generator, 4 GPIO (input or output) under software control allow command for security or control to low frequency devices.

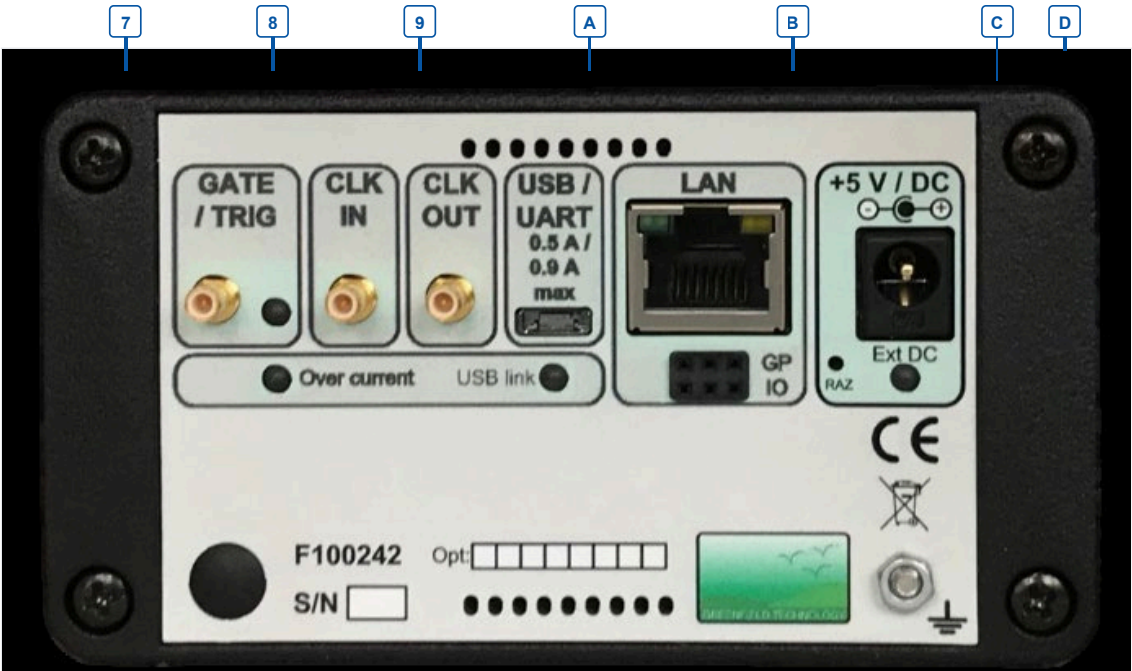


# Front and Rear Panel

Front Panel



Rear Panel



Connectors, Switch

|   | Front Panel                              |   | Rear Panel                         |
|---|------------------------------------------|---|------------------------------------|
| 1 | Trigger input: SMB connector             | 7 | Gate input: SMB connector          |
| 2 | T1 channel output: SMB connector         | 8 | Clock input: SMB connector         |
| 3 | T2 channel output: SMB connector         | 9 | Clock output: SMB connector        |
| 4 | T3 channel output: SMB connector         | A | USB connection: micro-AB connector |
| 5 | T4 channel output: SMB connector         | B | LAN connection: RJ45 connector     |
| 6 | Power ON/OFF or RUN/STOP channel outputs | C | GPIO: SHM-103 Samtec connector     |
|   |                                          |   | +5V DC power plug: Jack 2.10 mm    |

# Pulse Shaping Modules

| Function                        | Description                                                                                    |
|---------------------------------|------------------------------------------------------------------------------------------------|
| Optical-to-electrical converter | Combined with the electrical-to-optical converter, allows transmitting a fast pulse up to 1 km |
| Sub-nanosecond pulse stretcher  | From pickup diode, to provide the Model 750 clock reference                                    |
| Line driver                     | 1 to 4 line, 50 $\Omega$ , up to 150 MHz rate                                                  |
| HV pulse generator              | 15–70 V, <2 ns rise time into 50 $\Omega$                                                      |
| TTL line driver                 | 4 channel, 50 $\Omega$ , up to 150 MHz rate                                                    |

## Specifications

| Delay Channels                               |                                                                                                                                                                               |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number                                       | 4 independent (or 8 in option)                                                                                                                                                |
| Range                                        | 100 seconds                                                                                                                                                                   |
| Resolution                                   | 100 ps (1 ps in option)                                                                                                                                                       |
| RMS jitter                                   | <50 ps + delay $\times 10^{-7}$ , channel to channel; <1 ns, external trigger to any channel; <50 ps, external clock to any channel (10 ps with option 2)                     |
| Accuracy                                     | <250 ps + delay $\times 10^{-6}$ (<100 ps + delay $\times 10^{-6}$ with option 2)                                                                                             |
| Time base                                    | Internal 200 MHz, $\pm 1$ ppm stability                                                                                                                                       |
| External Trigger Mode                        |                                                                                                                                                                               |
| Input "TRIG"                                 | Rate single or repetitive up to 40 MHz, with prescaler, adjustable threshold from +100 mV to +5 V (step 1 mV), adjustable positive or negative slope and 5 ns min pulse width |
| Trigger delay                                | <85 ns (insertion delay)                                                                                                                                                      |
| Internal Trigger Mode                        |                                                                                                                                                                               |
| Rate, repetitive                             | From three timers with frequency = 0.01 Hz to 50 MHz (in steps of 5 ns)                                                                                                       |
| Rate, single                                 | Synchronized on internal timers or software command                                                                                                                           |
| Channel Output Pulse T1 to T4 (and T5 to T8) |                                                                                                                                                                               |
| Amplitude                                    | 1.5 V to 5 V in 1 mV steps into 50 $\Omega$ ; 3.0 V to 10 V in 2 mV steps into high impedance (>1 k $\Omega$ )                                                                |
| Rise/fall time                               | 1 ns / 1 ns into 50 $\Omega$ , or 2 ns / 2 ns into high impedance at full scale                                                                                               |
| Width                                        | 10 ns to 10 s in 5 ns steps                                                                                                                                                   |
| Pulse polarity                               | Positive or negative                                                                                                                                                          |
| Burst mode                                   | Burst count = 1 to 1,000,000,000, adjustable period in 5 ns steps                                                                                                             |
| MUX mode                                     | Any channel may be OR'ed to all outputs (per group of 4)                                                                                                                      |



| External Clock Reference                 |                                                                                                    |
|------------------------------------------|----------------------------------------------------------------------------------------------------|
| Threshold Level                          | 0 V, internal 50 $\Omega$                                                                          |
| Frequency                                | Min -10 dBm, typical 6 dBm                                                                         |
|                                          | 10 MHz to 240 MHz, user programmable in steps of 0.25 MHz up to 120 MHz then in steps of 0.50 MHz  |
| Clock Output                             |                                                                                                    |
| Level                                    | >500 mV p-p, into 50 $\Omega$ , AC coupling                                                        |
| Frequency                                | 100 MHz if internally clocked, or same as external if externally clocked (800 MHz with option 2)   |
| Gate or Second Trigger or Daisy Chaining |                                                                                                    |
| Function                                 | Gate or second external trigger with adjustable threshold, or daisy chaining to a second Model 750 |
| Input                                    | Active high or low in gate mode, positive/negative slope with rate <40 MHz in trigger mode         |
| GPIO: Input or Output Lines              |                                                                                                    |
| 4 $\times$ GPIO                          | Input or output, 0 or 3 V level, impedance >20 k $\Omega$ input & 100 $\Omega$ output              |
| General                                  |                                                                                                    |
| Interface control                        | USB to UART, Ethernet 10/100 Mb/s                                                                  |
| User's sets                              | 8 sets to save or recall eight specific configurations                                             |
| Software tools                           | Free drivers for Windows 11 and Linux; USB application with GUI included                           |
| Power consumption                        | 2.5 W to 15 W according to configuration                                                           |
| Power supply                             | USB or external AC (80–264 V / 47–63 Hz) to DC (5 V, 4 A), adapter included                        |
| Size and weight                          | 108 $\times$ 58.6 $\times$ 129 mm, <1 kg                                                           |

## Options

**Option 1:** Extension to 8 channels

**Option 2:** 1 ps delay resolution (only available with Model 750-4C), with channel-to-channel jitter <10 ps (at short delay) and min width of 50 ns. Narrow pulse mode: min width of 3 ns in 1 ps steps on 2 channel outputs

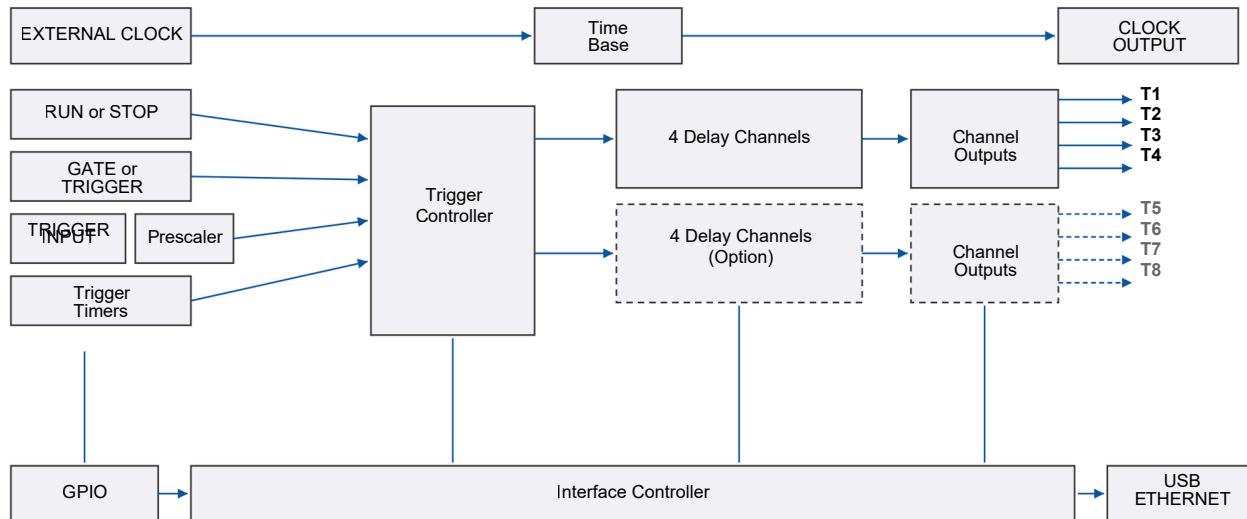
**Option 9:** 5 inchSMB-to-BNC cable

## Ordering Information

| Model        | Description                                            |
|--------------|--------------------------------------------------------|
| Model 750-4C | 4 channel mini pulse & delay generator (base version)  |
| Model 750-8C | 8 channel mini pulse & delay generator (with option 1) |

# Operating Information

## Block Diagram of the Generator



**Time base:** This function provides a 200 MHz time base from an internal reference or an external 10 MHz to 240 MHz reference. The internal time base is available (Clock out) on the rear panel.

**Trigger controller:** This function provides 2 trigger modes. *External Trigger Mode:* a rising edge on input "Trigger input" triggers all delay channels; every channel's trigger rate can be single, repetitive, or inhibited, and a prescaler may divide the Trigger Input frequency by an integer from 1 to 1,000,000,000. *Internal Trigger Mode:* delay channels can be triggered from 3 frequency-programmable timers; every channel's trigger rate can be single (synchronized on trigger sources), repetitive, burst, or inhibited.

The seven (7) trigger sources possible for each channel are presented below.

|               | Trigger input | Gate input | Timer N°1 | Timer N°2 | Timer N°3 | Command | Inhibited |
|---------------|---------------|------------|-----------|-----------|-----------|---------|-----------|
| External mode | X             | X          |           |           |           |         |           |
| Internal mode |               |            | X         | X         | X         | X       | X         |

Trigger sources

"Gate Input" allows quickly inhibiting all selected channel outputs. This input function can be selected as a second external trigger or an input/output for daisy chaining two Model 750 generators. The Run or Stop button enables or disables the output for any channel.

**Delay Channel:** 4 independent delay channels (or 8 in option). The delay from the selected trigger source is programmable up to 100 seconds in 100 ps increments (1 ps in option).

**Channel Output:** Each delayed output pulse (T1 to T4 or T5 to T8) can be independently adjustable in level (1.5 V to 5 V in 1 mV steps), width (10 ns to 10 s in 5 ns steps), and polarity, and may be OR'ed to all other outputs. The outputs are designed to drive a 50  $\Omega$  load; on a high-impedance load, output level will be twice.

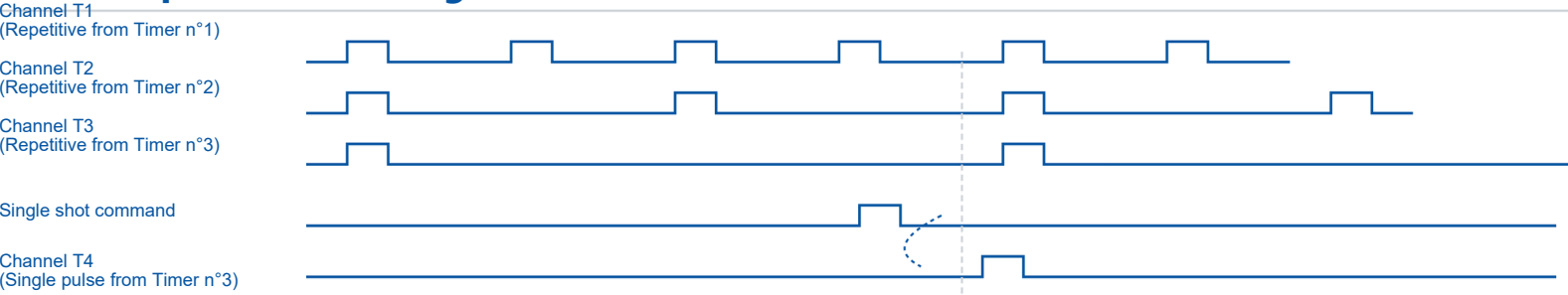
**Burst mode:** On each channel output, the number of pulses is adjustable with burst count and burst period. **Narrow pulse mode** (option 2): uses two delay channels, one to start and one to stop the output pulse, for a narrower, precisely adjusted pulse.

**Interface Controller:** Manages internal functions and the user interface. All parameters can be remote controlled via USB to UART and Ethernet (10/100 Mb/s), and are automatically saved across 8 different user sets. **GPIO:** four lines, input or output, under software command, allow control of other devices.

**Stand-alone:** Model 750 can be used without a PC; a PC is only needed to change the pulse characteristics. When powered up, an Auto-Run mode allows the unit to start providing output pulses automatically.

# Example of Channel Outputs Mode

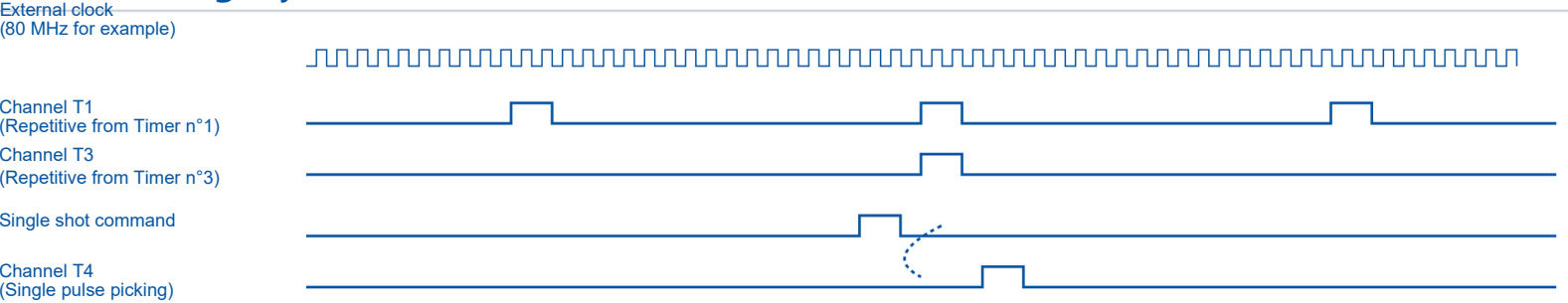
## Mode Repetitive and Single



## Mode Burst and Mux



## Pulse Picking (Synchronized on Timer N°3)





# Control and Software Tools

There are two ways to control the generator: Easy remote and General remote.

## Easy Remote (Web Control Panel)

Via Ethernet and control panel web pages, without any specific software application. The web page, from the embedded web server, provides an easy method to configure settings.

A Main menu allows display and control of: trigger and clock system (level, slope, prescaler, clock input/output, trigger generators F1/F2/F3); delay channel (trigger source, rate, delay, amplitude/width/polarity); extended delay channel settings (burst mode, gate, MUX mode); a "RUN" button enabling output for all delay channels; a "Trigger" button for a single trigger synchronized with the selected trigger source

A secondary menu ("parameter") allows changing the IP address and configuring specific parameters

Another secondary menu ("GPIO") allows configuring input/output lines

"Chart View" opens a window displaying every channel output on a common time chart in real time, with zoom and cursor functions for time detail

The configuration information (all settings) of the instrument is stored and saved in the Model 750. The web page can be opened via Edge, Mozilla Firefox, or Chrome. After connecting a cable from the Model 750 Ethernet port to your computer network, enter the Model 750 IP address into your PC's browser (identified in the User's Manual); the browser will automatically open the control panel web page.

**TRIGGER and CLOCK SYSTEM**

|       |         |           |         |      |
|-------|---------|-----------|---------|------|
| Input | Type    | Prescaler | Level   | Edge |
| Ext1  | Trigger | 1         | 0 mV    |      |
| Ext2  | Trigger | 1         | 1000 mV |      |

External Clock No

External Clock Frequency 80 000 000 Hz

Clock Output On

Output Frequency 100 000 000 Hz

Generator F1 1 000.000 F2 50.000 F3 1.000 Hz

RUN

Trigger

**DELAY CHANNEL** Chart View

|           | Trigger Source | Trigger Rate | Delay ps (step=100ps) | Width ns | Amplitude mV | Polarity |
|-----------|----------------|--------------|-----------------------|----------|--------------|----------|
| T1 ( 5V)  | Generator F1   | Repetitive   | 2 000 000             | 100      | 5 000        | Positive |
| T2 ( 5V)  | Generator F1   | Repetitive   | 1 000 000             | 50       | 2 500        | Negative |
| T3 ( 10V) | Generator F1   | Repetitive   | 12 000 000            | 2 000    | 10 000       |          |
| T4 ( 10V) | Generator F1   | Repetitive   | 500 000               | 10 000   | 9 500        |          |

**EXTENDED SETUP**

Temperature 40.5 °C

Current 628 mA

CLKIN Frequency 0 Hz

Lock ☒ External Power ☒ Fault ☒ Overload ☒

Save/Recall

Settings

GPIO

Info